

Data Base Management System (DBMS)

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Roll No : 15

Batch : B1

Assignment No 1 - DDL and DML Commands

Question 1: Employee Management

Table Structure:

Employee(EmpID, EmpName, Department, Salary)

Perform the following SQL operations:

DDL Commands:

a) Create the Employee table with appropriate data types.

```
mysql> create table Employee(EmpID int,EmpName varchar(50),Department varchar(50),Salary int);
Query OK, 0 rows affected (0.04 sec)

mysql> show tables;
+-----+
| Tables_in_Harshad |
+-----+
| Employee           |
+-----+
1 row in set (0.01 sec)

mysql> desc Employee;
+-----+-----+-----+-----+-----+-----+
| Field      | Type          | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| EmpID      | int           | YES  |     | NULL    |       |
| EmpName    | varchar(50)   | YES  |     | NULL    |       |
| Department | varchar(50)   | YES  |     | NULL    |       |
| Salary     | int           | YES  |     | NULL    |       |
+-----+-----+-----+-----+-----+-----+
4 rows in set (0.01 sec)
```

b) Add a new column City.

```
mysql> alter table Employee add City varchar(35);
Query OK, 0 rows affected (0.04 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> desc Employee;
+-----+-----+-----+-----+-----+-----+
| Field      | Type          | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| EmpID      | int           | YES  |     | NULL    |       |
| EmpName    | varchar(50)   | YES  |     | NULL    |       |
| Department | varchar(50)   | YES  |     | NULL    |       |
| Salary     | int           | YES  |     | NULL    |       |
| City       | varchar(35)   | YES  |     | NULL    |       |
+-----+-----+-----+-----+-----+-----+
5 rows in set (0.00 sec)
```

c) Add a new column JoiningDate.

```
mysql> alter table Employee add JoiningDate Date;;
Query OK, 0 rows affected (0.06 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> desc Employee;
+-----+-----+-----+-----+-----+-----+
| Field      | Type          | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| EmpID      | int           | YES  |     | NULL    |       |
| EmpName    | varchar(50)   | YES  |     | NULL    |       |
| Department | varchar(50)   | YES  |     | NULL    |       |
| Salary     | int           | YES  |     | NULL    |       |
| City       | varchar(35)   | YES  |     | NULL    |       |
| JoiningDate | date          | YES  |     | NULL    |       |
+-----+-----+-----+-----+-----+-----+
6 rows in set (0.00 sec)
```

d) Modify the size of the Department column.

```
mysql> alter table Employee modify Department Varchar(30);
Query OK, 0 rows affected (0.06 sec)
Records: 0 Duplicates: 0 Warnings: 0

mysql> desc Employee;
```

Field	Type	Null	Key	Default	Extra
EmpID	int	YES		NULL	
EmpName	varchar(30)	YES		NULL	
Department	varchar(30)	YES		NULL	
Salary	float	YES		NULL	
City	varchar(35)	YES		NULL	
JoiningDate	date	YES		NULL	

6 rows in set (0.00 sec)

e) Modify the size of the EmpName column.

```
mysql> alter table Employee modify EmpName Varchar(30);
Query OK, 0 rows affected (0.08 sec)
Records: 0 Duplicates: 0 Warnings: 0

mysql> desc Employee;
```

Field	Type	Null	Key	Default	Extra
EmpID	int	YES		NULL	
EmpName	varchar(30)	YES		NULL	
Department	varchar(50)	YES		NULL	
Salary	float	YES		NULL	
City	varchar(35)	YES		NULL	
JoiningDate	date	YES		NULL	

6 rows in set (0.00 sec)

f) Rename the column Salary to MonthlySalary.

```
mysql> alter table Employee change Salary MonthlySalary float;
Query OK, 0 rows affected (0.02 sec)
Records: 0 Duplicates: 0 Warnings: 0

mysql> desc Employee;
```

Field	Type	Null	Key	Default	Extra
EmpID	int	YES		NULL	
EmpName	varchar(30)	YES		NULL	
Department	varchar(30)	YES		NULL	
MonthlySalary	float	YES		NULL	
City	varchar(35)	YES		NULL	
JoiningDate	date	YES		NULL	

```
6 rows in set (0.00 sec)
```

g) Drop the JoiningDate column.

```
mysql> alter table Employee drop joiningDate;
Query OK, 0 rows affected (0.05 sec)
Records: 0 Duplicates: 0 Warnings: 0

mysql> desc Employee;
```

Field	Type	Null	Key	Default	Extra
EmpID	int	YES		NULL	
EmpName	varchar(30)	YES		NULL	
Department	varchar(30)	YES		NULL	
MonthlySalary	float(8,2)	YES		NULL	
City	varchar(35)	YES		NULL	

```
5 rows in set (0.01 sec)
```

h) Rename the table Employee to EmployeeDetails.

```
mysql> rename table Employee to EmployeeDetails;
Query OK, 0 rows affected (0.01 sec)

mysql> select * from EmployeeDetails;
Empty set (0.00 sec)
```

DML Commands:

i) Insert at least 5 employee records into the table.

```
mysql> insert into EmployeeDetails(EmpID,EmpName,Department,MonthlySalary,City)value(1,'Harshad Teli','Testing',40000,'Kolhapur');
Query OK, 1 row affected (0.01 sec)

mysql> select * from EmployeeDetails;
+-----+-----+-----+-----+-----+
| EmpID | EmpName      | Department | MonthlySalary | City      |
+-----+-----+-----+-----+-----+
| 1     | Harshad Teli | Testing    | 40000.00      | Kolhapur |
+-----+-----+-----+-----+-----+
1 row in set (0.00 sec)
```

j) Display all records from the table.

```
mysql> select * from EmployeeDetails;
+-----+-----+-----+-----+-----+
| EmpID | EmpName      | Department      | MonthlySalary | City      |
+-----+-----+-----+-----+-----+
| 1     | Harshad Teli | Testing         | 40000.00      | Kolhapur |
| 2     | Anurag Lad   | Marketing       | 350000.00     | Pune      |
| 3     | Vardhan Patki | designer        | 25000.00      | Satara    |
| 4     | Siddhi Zure  | Sell Management | 20000.00      | Ishwarpur |
| 5     | Prajkta Patil | Stock Management | 20000.00      | Mumbai    |
+-----+-----+-----+-----+-----+
5 rows in set (0.00 sec)
```

k) Update the City of an employee using EmpID.

```
mysql> update EmployeeDetails set City='Karad' where EmpID=2;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1 Changed: 1 Warnings: 0

mysql> select * from EmployeeDetails;
```

EmpID	EmpName	Department	MonthlySalary	City
1	Harshad Teli	Testing	40000.00	Kolhapur
2	Anurag Lad	Marketing	35000.00	Karad
3	Vardhan Patki	designer	25000.00	Satara
4	Siddhi Zure	Sell Management	20000.00	Ishwarpur
5	Prajakta Patil	Stock Management	20000.00	Mumbai

```
5 rows in set (0.00 sec)
```

l) Update the MonthlySalary of an employee using EmpID.

```
mysql> update EmployeeDetails set MonthlySalary = 35000 where EmpID = 2;
Query OK, 1 row affected (0.02 sec)
Rows matched: 1 Changed: 1 Warnings: 0

mysql> select * from EmployeeDetails;
```

EmpID	EmpName	Department	MonthlySalary	City
1	Harshad Teli	Testing	40000.00	Kolhapur
2	Anurag Lad	Marketing	35000.00	Karad
3	Vardhan Patki	designer	25000.00	Satara
4	Siddhi Zure	Sell Management	20000.00	Ishwarpur
5	Prajakta Patil	Stock Management	20000.00	Mumbai

```
5 rows in set (0.00 sec)
```

m) Update the Department of an employee using EmpID.

```
mysql> update EmployeeDetails set Department = 'Marketing' where EmpID = 5;
Query OK, 1 row affected (0.03 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> select * from EmployeeDetails;
+-----+-----+-----+-----+-----+
| EmpID | EmpName      | Department      | MonthlySalary | City      |
+-----+-----+-----+-----+-----+
| 1     | Harshad Teli | Testing         | 40000.00      | Kolhapur  |
| 2     | Anurag Lad   | Marketing       | 35000.00      | Karad     |
| 3     | Vardhan Patki | designer        | 25000.00      | Satara    |
| 4     | Siddhi Zure  | Sell Management | 20000.00      | Ishwarpur |
| 5     | Prajkta Patil | Marketing       | 20000.00      | Mumbai    |
+-----+-----+-----+-----+-----+
5 rows in set (0.00 sec)
```

n) Delete one employee record using EmpID.

```
mysql> delete from EmployeeDetails where EmpID = 4;
Query OK, 1 row affected (0.02 sec)

mysql> select * from EmployeeDetails;
+-----+-----+-----+-----+-----+
| EmpID | EmpName      | Department      | MonthlySalary | City      |
+-----+-----+-----+-----+-----+
| 1     | Harshad Teli | Testing         | 40000.00      | Kolhapur  |
| 2     | Anurag Lad   | Marketing       | 35000.00      | Karad     |
| 3     | Vardhan Patki | designer        | 25000.00      | Satara    |
| 5     | Prajkta Patil | Marketing       | 20000.00      | Mumbai    |
+-----+-----+-----+-----+-----+
4 rows in set (0.00 sec)
```

o) Display the records after performing the update and delete operations.

```
mysql> select * from EmployeeDetails;
+-----+-----+-----+-----+-----+
| EmpID | EmpName      | Department      | MonthlySalary | City      |
+-----+-----+-----+-----+-----+
| 1     | Harshad Teli | Testing         | 40000.00      | Kolhapur  |
| 2     | Anurag Lad   | Marketing       | 350000.00     | Pune      |
| 3     | Vardhan Patki | designer        | 25000.00      | Satara    |
| 4     | Siddhi Zure  | Sell Management | 20000.00      | Ishwarpur |
| 5     | Prajkta Patile | Stock Management | 20000.00      | Mumbai    |
+-----+-----+-----+-----+-----+
5 rows in set (0.00 sec)
```

Question 2: Student Management

Table Structure:

Student(StudentID, StudentName, Course, Age)

Perform the following SQL operations:

DDL Commands:

a) Create the Student table with appropriate data types.

```
mysql> Create table Student(StudentID int,StudentName Varchar(50),Course Varchar(50),Age int);
Query OK, 0 rows affected (0.03 sec)

mysql> desid Student;
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near 'Student' at line 1
mysql> desc Student;
+-----+-----+-----+-----+-----+-----+
| Field      | Type          | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| StudentID  | int           | YES  |     | NULL    |       |
| StudentName | varchar(50)    | YES  |     | NULL    |       |
| Course     | varchar(50)    | YES  |     | NULL    |       |
| Age        | int           | YES  |     | NULL    |       |
+-----+-----+-----+-----+-----+-----+
4 rows in set (0.00 sec)
```

b) Add a new column Email.

```
mysql> alter table Student Add Email Varchar(20);
Query OK, 0 rows affected (0.04 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> desc Student;
+-----+-----+-----+-----+-----+-----+
| Field      | Type          | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| StudentID  | int           | YES  |     | NULL    |       |
| StudentName | varchar(50)    | YES  |     | NULL    |       |
| Course     | varchar(50)    | YES  |     | NULL    |       |
| Age        | int           | YES  |     | NULL    |       |
| Email      | varchar(20)    | YES  |     | NULL    |       |
+-----+-----+-----+-----+-----+-----+
5 rows in set (0.00 sec)
```


c) Add a new column MobileNo.

```
mysql> desc Student;
```

Field	Type	Null	Key	Default	Extra
StudentID	int	YES		NULL	
StudentName	varchar(50)	YES		NULL	
Course	varchar(50)	YES		NULL	
Age	int	YES		NULL	
Email	varchar(20)	YES		NULL	
MobileNo	int	YES		NULL	

6 rows in set (0.00 sec)

d) Modify the size of the Course column.

```
mysql> alter table Student modify Course varchar(30);
Query OK, 0 rows affected (0.07 sec)
Records: 0 Duplicates: 0 Warnings: 0

mysql> desc Student;
```

Field	Type	Null	Key	Default	Extra
StudentID	int	YES		NULL	
StudentName	varchar(50)	YES		NULL	
Course	varchar(30)	YES		NULL	
Age	int	YES		NULL	
Email	varchar(20)	YES		NULL	
MobileNo	int	YES		NULL	

6 rows in set (0.00 sec)

e) Modify the size of the StudentName column.

```
mysql> alter table Student modify StudentName varchar(40);;
Query OK, 0 rows affected (0.06 sec)
Records: 0 Duplicates: 0 Warnings: 0

ERROR:
No query specified

mysql> desc Student;
```

Field	Type	Null	Key	Default	Extra
StudentID	int	YES		NULL	
StudentName	varchar(40)	YES		NULL	
Course	varchar(30)	YES		NULL	
Age	int	YES		NULL	
Email	varchar(20)	YES		NULL	
MobileNo	int	YES		NULL	

```
6 rows in set (0.01 sec)
```

f) Rename the column Age to StudentAge.

```
mysql> alter table Student change Age StudentAge int;
Query OK, 0 rows affected (0.01 sec)
Records: 0 Duplicates: 0 Warnings: 0

mysql> desc Student;
```

Field	Type	Null	Key	Default	Extra
StudentID	int	YES		NULL	
StudentName	varchar(40)	YES		NULL	
Course	varchar(30)	YES		NULL	
StudentAge	int	YES		NULL	
Email	varchar(20)	YES		NULL	
MobileNo	int	YES		NULL	

```
6 rows in set (0.00 sec)
```

g) Drop the Email column

```
mysql> alter table Student drop email;
Query OK, 0 rows affected (0.07 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> desc Student;
+-----+-----+-----+-----+-----+-----+
| Field          | Type          | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| StudentID      | int           | YES  |     | NULL    |       |
| StudentName    | varchar(40)   | YES  |     | NULL    |       |
| Course         | varchar(30)   | YES  |     | NULL    |       |
| StudentAge     | int           | YES  |     | NULL    |       |
| MobileNo       | int           | YES  |     | NULL    |       |
+-----+-----+-----+-----+-----+-----+
5 rows in set (0.00 sec)
```

h) Rename the table Student to StudentMaster.

```
mysql> rename table Student to StudentMaster;
Query OK, 0 rows affected (0.02 sec)

mysql> desc StudentMaster;
+-----+-----+-----+-----+-----+-----+
| Field          | Type          | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| StudentID      | int           | YES  |     | NULL    |       |
| StudentName    | varchar(40)   | YES  |     | NULL    |       |
| Course         | varchar(30)   | YES  |     | NULL    |       |
| StudentAge     | int           | YES  |     | NULL    |       |
| MobileNo       | int           | YES  |     | NULL    |       |
+-----+-----+-----+-----+-----+-----+
5 rows in set (0.00 sec)
```

DML Commands:

i) Insert at least 5 student records into the table.

```
mysql> insert into StudentMaster(StudentID,StudentName,Course,StudentAge,MobileNo)value(1,'Harshad Teli','BCS',18,9970898012);
Query OK, 1 row affected (0.01 sec)

mysql> select * from StudentMaster;
+-----+-----+-----+-----+-----+
| StudentID | StudentName | Course | StudentAge | MobileNo |
+-----+-----+-----+-----+-----+
| 1 | Harshad Teli | BCS | 18 | 9970898012 |
+-----+-----+-----+-----+-----+
1 row in set (0.00 sec)

mysql> insert into StudentMaster(StudentID,StudentName,Course,StudentAge,MobileNo)values(2,'Anurag Patil','BCA',18,9156371425);
Query OK, 1 row affected (0.02 sec)

mysql> insert into StudentMaster(StudentID,StudentName,Course,StudentAge,MobileNo)values(3,'Vardhan Patki','BCOM',20,7029536728);
Query OK, 1 row affected (0.02 sec)

mysql> insert into StudentMaster(StudentID,StudentName,Course,StudentAge,MobileNo)values(4,'Rajveer Patil','BA',21,7029536728);
Query OK, 1 row affected (0.01 sec)

mysql> insert into StudentMaster(StudentID,StudentName,Course,StudentAge,MobileNo)values(5,'Shruti Patili','BSC',23,9970898012);
Query OK, 1 row affected (0.03 sec)

mysql> select * from StudentMaster;
+-----+-----+-----+-----+-----+
| StudentID | StudentName | Course | StudentAge | MobileNo |
+-----+-----+-----+-----+-----+
| 1 | Harshad Teli | BCS | 18 | 9970898012 |
| 2 | Anurag Patil | BCA | 18 | 9156371425 |
| 3 | Vardhan Patki | BCOM | 20 | 7029536728 |
| 4 | Rajveer Patil | BA | 21 | 7029536728 |
| 5 | Shruti Patili | BSC | 23 | 9970898012 |
+-----+-----+-----+-----+-----+
5 rows in set (0.00 sec)
```

j) Display all records from the table.

```
mysql> select * from StudentMaster;
+-----+-----+-----+-----+-----+
| StudentID | StudentName | Course | StudentAge | MobileNo |
+-----+-----+-----+-----+-----+
| 1 | Harshad Teli | BCS | 18 | 9970898012 |
| 2 | Anurag Patil | BCA | 18 | 9156371425 |
| 3 | Vardhan Patki | BCOM | 20 | 7029536728 |
| 4 | Rajveer Patil | BA | 21 | 7894561245 |
| 5 | Shruti Patili | BSC | 23 | 9970898012 |
+-----+-----+-----+-----+-----+
5 rows in set (0.00 sec)
```

k) Update the Course of a student using StudentID.

```
mysql> update StudentMaster set Course = 'B.Tech' where StudentID=4;
Query OK, 1 row affected (0.02 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> select * from StudentMaster;
+-----+-----+-----+-----+-----+
| StudentID | StudentName | Course | StudentAge | MobileNo |
+-----+-----+-----+-----+-----+
| 1 | Harshad Teli | BCS | 18 | 9970898012 |
| 2 | Anurag Patil | BCA | 18 | 9156371425 |
| 3 | Vardhan Patki | BCOM | 20 | 7029536728 |
| 4 | Rajveer Patil | B.Tech | 21 | 7894561245 |
| 5 | Shruti Patili | BSC | 23 | 9970898012 |
+-----+-----+-----+-----+-----+
5 rows in set (0.00 sec)
```

l) Update the StudentAge of a student using StudentID.

```
mysql> update StudentMaster set StudentAge = 22 where StudentID = 2;
Query OK, 1 row affected (0.02 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> select * from StudentMaster;
+-----+-----+-----+-----+-----+
| StudentID | StudentName | Course | StudentAge | MobileNo |
+-----+-----+-----+-----+-----+
| 1 | Harshad Teli | BCS | 18 | 9970898012 |
| 2 | Anurag Patil | BCA | 22 | 9156371425 |
| 3 | Vardhan Patki | BCOM | 20 | 7029536728 |
| 4 | Rajveer Patil | B.Tech | 21 | 7894561245 |
| 5 | Shruti Patili | BSC | 23 | 9970898012 |
+-----+-----+-----+-----+-----+
5 rows in set (0.00 sec)
```

m) Update the MobileNo of a student using StudentID.

```
mysql> update StudentMaster set MobileNo= 7894561245 where StudentID=4;
Query OK, 1 row affected (0.03 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> select * from StudentMaster;
+-----+-----+-----+-----+-----+
| StudentID | StudentName | Course | StudentAge | MobileNo |
+-----+-----+-----+-----+-----+
| 1 | Harshad Teli | BCS | 18 | 9970898012 |
| 2 | Anurag Patil | BCA | 18 | 9156371425 |
| 3 | Vardhan Patki | BCOM | 20 | 7029536728 |
| 4 | Rajveer Patil | BA | 21 | 7894561245 |
| 5 | Shruti Patili | BSC | 23 | 9970898012 |
+-----+-----+-----+-----+-----+
5 rows in set (0.00 sec)
```

n) Delete one student record using StudentID.

```
mysql> delete from StudentMaster where StudentID = 3;
Query OK, 1 row affected (0.02 sec)

mysql> select * from StudentMaster;
+-----+-----+-----+-----+-----+
| StudentID | StudentName | Course | StudentAge | MobileNo |
+-----+-----+-----+-----+-----+
| 1 | Harshad Teli | BCS | 18 | 9970898012 |
| 2 | Anurag Patil | BCA | 22 | 9156371425 |
| 4 | Rajveer Patil | B.Tech | 21 | 7894561245 |
| 5 | Shruti Patili | BSC | 23 | 9970898012 |
+-----+-----+-----+-----+-----+
4 rows in set (0.00 sec)
```

o) Display the records after performing the update and delete operations.

```
mysql> select * from StudentMaster;
+-----+-----+-----+-----+-----+
| StudentID | StudentName | Course | StudentAge | MobileNo |
+-----+-----+-----+-----+-----+
| 1 | Harshad Teli | BCS | 18 | 9970898012 |
| 2 | Anurag Patil | BCA | 22 | 9156371425 |
| 4 | Rajveer Patil | B.Tech | 21 | 7894561245 |
| 5 | Shruti Patili | BSC | 23 | 9970898012 |
+-----+-----+-----+-----+-----+
4 rows in set (0.00 sec)
```

